Improving Matches for a Dating Site

Decision Tree, KNN, and Naive Bayes

Abdullah Mohammed

Ontario Tech University

CSCI 4150U: Data Mining

Lab 5

March 10, 2026

1 Dataset Description

Let $x = (x_1, x_2, x_3)$ denote the feature vector for one customer:

$$x_1 := \text{Frequent flyer miles per year}$$
$$x_2 := \text{Percent time playing video games}$$
$$x_3 := \text{Liters of ice cream per week}$$

The target label y is one of three preference categories:

$$y \in \{\texttt{didntLike}, \texttt{smallDoses}, \texttt{largeDoses}\}.$$

2 Methodology

Because the features have different scales, scaling is important for models such as KNN and Naive-Bayes. In this lab, I used:

- **Standardization** (z-score scaling) for KNN and Naive Bayes:

$$z = \frac{x - \mu}{\sigma}$$

- **No scaling required** for Decision Trees because splits depend on thresholds, not distances.

2.1 Models

- **Decision Tree**
- **KNN**
- **Naive Bayes**

2.2 Evaluation Methodologies

2.2.1 Held-out Test Set

Models are trained using `datingData_training.txt` and evaluated once on `datingData_test.txt`.

2.2.2 Stratified k-Fold Cross-Validation

I used **Stratified 5-fold cross-validation** on the training set.

3 Performance Metrics

- Accuracy
- Precision
- F-Measure

Table 1: Stratified 5-Fold Cross-Validation Performance (Training Set)

Model	Accuracy	Precision	F-measure
Decision Tree	0.9383	0.9398	0.9388
KNN	0.9483	0.9500	0.9487
Naive Bayes	0.9350	0.9375	0.9358

4 Results

4.1 Cross-Validation Results

Table 1 reports mean across the 5 folds.

4.2 Test Set Results

Table 2 reports final held-out test performance.

Table 2: Held-out Test Set Performance

Model	Accuracy	Precision	F-measure
Decision Tree	0.9575	0.9569413018003622	0.9573954599761052
KNN	0.96	0.9604466316642927	.9599889735722344
Naive Bayes	0.93	0.9306551657297927	0.9272542055353591

4.3 Learning Curves (Training vs Validation Accuracy)

Learning curves show how training and validation accuracy change as the training set size increases.

Insert figures exported from the notebook (recommended file names shown below):

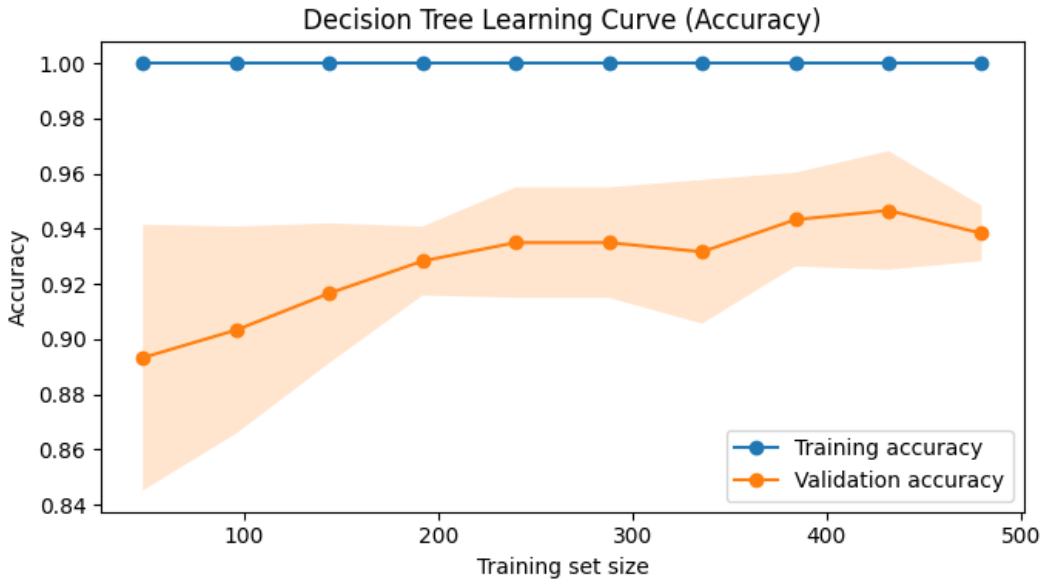


Figure 1: Decision Tree learning curve (accuracy vs. training size).

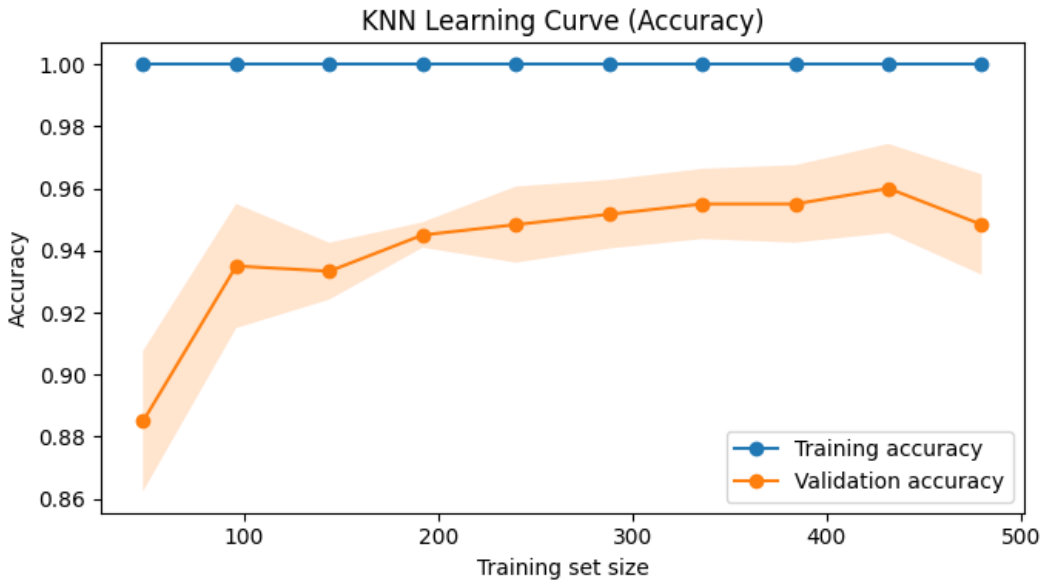


Figure 2: KNN learning curve (accuracy vs. training size).

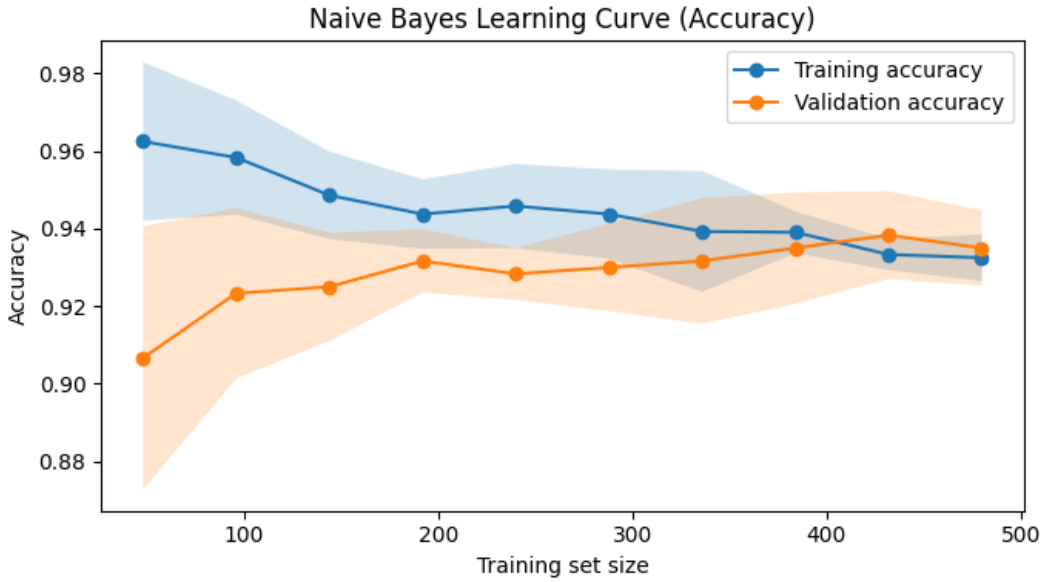


Figure 3: Naive Bayes learning curve (accuracy vs. training size).

4.4 Decision Tree Graph

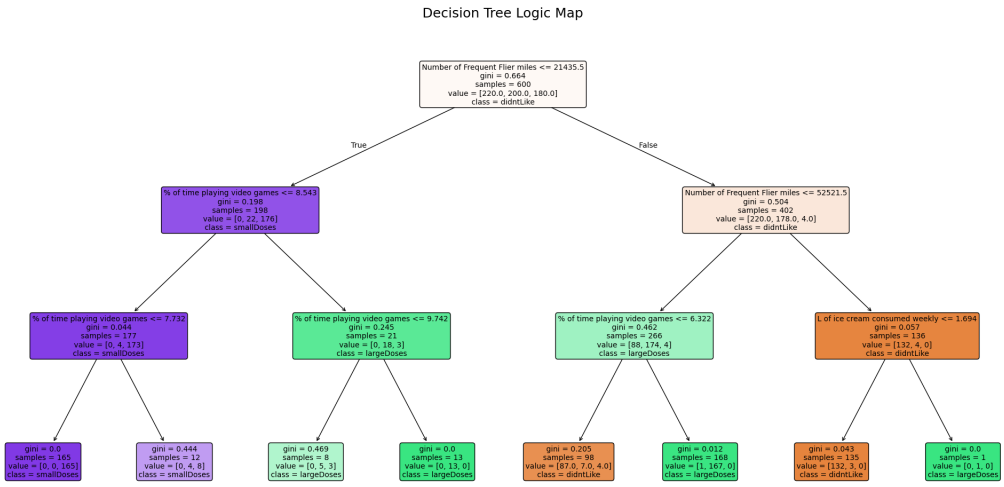


Figure 4: Figure 4 shows that Decision Tree model that was chosen. In this model, the green boxes represent largeDoses,the purple boxes represent smallDoses, while the orange boxes represent didntLike.

5 Discussion

5.1 Model Comparison

- **Decision Tree** as we can see from the graph above is overfitting. This is due to the training accuracy being 1 throughout the entire training process
- **KNN** is also overfitting due to the reasons stated above.
- **Naive Bayes** is not overfitting and is actually showing that it is learning trends. This is due to the validation accuracy being similar to the accuracy of the training data.

6 Possible Solutions

Based on our performance and learning curve analyses, here are three suggestions to improve the dating site.

Suggestion 1: Using the Naive Bayes model for the matching algorithm

Our analysis showed that while the Decision Tree and KNN had higher accuracy, the graphs above show that it is overfitting. This is due to the fact that the accuracy of the training data stays at 1(100%) while the validation data accuracy slightly increases but does not converge with the training accuracy. However, the Naive Bayes model is not overfitting. This is why the dating site should use a Naive Bayes model for matching people.

Suggestion 2: Use the findings from the Decision Tree graph to make the algorithm better

Figure 4 shows the Decision Tree. We can use the findings from that Decision Tree to make a better matching algorithm. For example, from the graph, we can glean that if the person has 'Number of Frequent Flier Miles' ¡= 21435.5 and '% of time playing video games' ¡= 9.742 then the classification is largeDoses. The Gini in this case is also 0 so that is a clear indication that people like that type of person frequently. Therefore, using the decision tree, we can fine-tune the algorithm to match potential partners better.

Suggestion 3: Track more user data to extract more features so that we can improve the model even further

While our current dataset gives us a satisfactory of understanding user preferences, we can significantly boost the model's accuracy by capturing more data points. By extracting new features that capture a wider range of a user's lifestyle and personality, we can build a much stronger matchmaking algorithm. This would ultimately leading to better, more successful connections for our users.

7 Conclusion

This lab compared three multiclass classifiers on dating preference data using stratified cross-validation and a held-out test set. Performance was evaluated using accuracy, precision, F-measure, and learning curves were used to diagnose overfitting/underfitting behavior.